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Optical access network

Abstract

An object of the present invention is to provide an optical access network that has high reliability for supporting information communication services and efficiently responds to difficult-to-predict optical fiber demand. An optical access network configuration according to the present invention includes an optical fiber cable **11** that is looped to connect a wiring section **25** and an exchange office **10** together (hereinafter, an upper loop), an optical fiber cable **21** that is looped and laid in each wiring section **25** (hereinafter, a lower loop), and an optical fiber switching function unit **31** installed at a connection point **30** between the upper loop **11** and the lower loop **21**. The optical fiber switching function unit **31** is a wiring board or an optical switch which can be switched by connector connection. The wiring board or the optical switch may be remotely controllable.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a 371 U.S. National Phase of International Application No. PCT/JP2021/002780, filed on Jan. 27, 2021. The entire disclosure of the above application is incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a field of design techniques of optical fiber cable network configurations in access-based communication networks.

BACKGROUND ART

(3) An optical access network is a network of communication equipment where base facilities of telecommunications carriers (hereinafter, exchange offices) and multiple user bases are connected by optical fiber cables. In order to stably provide information communication services, the network requires high reliability, and also requires efficient response to the demand for optical fibers that are scattered over the surface. Since the construction of an optical network requires a large amount of equipment, and time, and incurs high costs, it is important to design and construct an appropriate network structure in advance.

(4) Optical access networks generally use a network configuration based on a star-shaped wiring topology (see NPL 1 for other details). In this network configuration, wiring sections each having about several hundred users as a unit are set, and the exchange office and the plurality of wiring sections are connected in a tree branch shape by an optical fiber cable. A number of required optical fibers calculated based on demand prediction at the time of design are fixedly allocated to each

wiring section, and optical fiber cables are extended in a tree branch shape in the wiring section and connected to each user base.

CITATION LIST

Non Patent Literature

(5) [NPL 1] YONEMOTO, SONOBE, KIGAMI, HASEGAWA, AND MIURA, “Optical Wiring Method for Mass Opening and Rapid Response,” NTT Technical Journal, Vol. 18, No. 12, pp. 48-52, 2006.

SUMMARY OF INVENTION

Technical Problem

(6) In a network configuration based on the conventional star-shaped topology, since a redundant path is not provided between an exchange office and a user base, there is a risk that a communication service will be interrupted if there is only one failure point accompanied by breakage. Further, since the optical fibers are fixedly allocated to each of the wiring sections and cable routes, even just one wiring section or cable route in which a demand for optical fiber exceeding the prediction at design time requires additional construction due to the switching and extension of the optical fibers. In order to reduce additional construction, it is necessary to lay more optical fibers than the predicted value in advance, leading to an increase in the creation cost and a decrease in the optical fiber use efficiency. That is, in the star-shaped network configuration, there are two problems that it is difficult to secure high reliability and that it is difficult to efficiently use an optical fiber with respect to demand fluctuations.

(7) In particular, predicting the demand for optical fibers for mobile communication base stations, which have been increasing in recent years, is difficult because of the number and positions of the demand affected by various influences, such as radio wave propagation environment or communication traffic conditions, besides the requirement, of the reliability for the optical fibers, increased by the spread of new generation mobile communication systems. Therefore, in the star-type network configuration, there is a possibility that it will become difficult to secure the required reliability and to efficiently use the optical fiber as mobile communication services advance and expanded in the future.

(8) The present invention has been made in view of the above circumstances, and an object of the present invention is to provide an optical access network which has high reliability for supporting information communication service and efficiently responds to an optical fiber demand which is difficult to predict.

Solution to Problem

(9) In order to achieve the above-mentioned object, the optical access network according to the present invention has a loop-type wiring topology, is composed of a plurality of loop-shaped optical fiber cables, and has a core fiber switching function at a connection point between loops.

(10) Specifically, the optical access network according to the present invention is an optical access network including an upper loop optical fiber cable which includes a plurality of fixed station lines and at least one common line and is laid on a loop-like path passing through a communication carrier base facility, and a lower loop optical fiber cable which includes at least one user line and at least one redundant line and is laid on a loop-like path passing through a connection point with the upper loop optical fiber cable and a user base facility, in which a fixed connection line for connecting the user lines and the fixed station lines in a 1:1 ratio, and an optical fiber switching function unit for exclusively connecting the redundant line and the common line are provided at the connection point, the number of the common lines is smaller than the total number of the redundant lines included in each of the plurality of lower loop optical fiber cables, and the optical fiber switching function unit switches connection between the redundant line and the common line according to demand fluctuation of the lower loop optical fiber cable.

(11) Since the optical access network is composed of a plurality of loop-shaped optical fiber cables, a redundant path can be secured, and reliability can be enhanced. Further, the optical access

network can make the optical fibers of a part of the upper loop optical fiber cables shared with a plurality of lower loop optical fiber cables, can switch connection according to demand, and can efficiently respond to the demand.
(12) Therefore, the present invention can provide an optical access network which has high reliability for supporting information communication service and efficiently responds to difficult-to-predict optical fiber demand.
(13) The optical access network according to the present invention preferably has the following conditions in order to reduce the frequency of construction and the like in response against the fluctuation of demand and to increase the availability.
(14) In the optical access network, the loop-like path of the upper loop optical fiber cable is a path in which an improvement rate γ of availability in a case of the upper loop optical fiber cable is maximized with respect to availability when there is one path from the communication carrier base facility to the connection point in a wiring section where the lower loop optical fiber cable is laid.
(15) Specifically, the improvement rate γ expressed by Expression C1, and the loop-like path of the upper loop optical fiber cable is r.sub.loop that maximizes the improvement rate γ,
[Math . C1]
(16)
$$= \frac{C_1 C_2 \sigma\theta r_{\text{loop}} (R^2 - r_{\text{loop}}^2)}{2 + C_1 C_2 \sigma\theta r_{\text{loop}} (R^2 - r_{\text{loop}}^2)} \quad \text{(C1)}$$
where an area shape constructed by the upper loop optical fiber cable is a fan shape having a center angle θ and a radius R with the communication carrier base facility as an origin, the r.sub.loop is a distance from the communication carrier base facility to the connection point, σ is a demand density, and C1 and C2 are constants.
(17) In the optical access network according to the present invention, the number of optical fibers connecting the redundant line and the optical fiber switching function unit is a number that converges the number of optical fibers F.sub.total included in the upper loop optical fiber cable to a minimum.
(18) Specifically, the optical fiber number F.sub.total is calculated by Expression C2:
[Math . C2]
(19)
$$F_{\text{total}} = \underset{i=1}{\overset{N}{.\text{Math.}}} (1 - a_i) f_i + F_c \quad \text{(C2)}$$
where N is the number of the lower loop optical fiber cables connected to the upper loop optical fiber cable, a.sub.i is the proportion of the number of the redundant lines among the optical fibers included in the i-th lower loop optical fiber cable (hereinafter, flexible optical fiber ratio), and f.sub.i is an optical fiber included in the i-th lower loop optical fiber cable, and Fc is the number of the common lines.
(20) Here, f.sub.i satisfies Expression C3, and Fc satisfies Expression C4.
[Math . C3]
(21)
$$\underset{k=0}{\overset{2f_i}{.\text{Math.}}} p_{i,k} \geq x \quad \text{(C3)}$$
(22) x is a design value of reliability per wiring section where the lower loop optical fiber cable is laid, and p.sub.i, k is a probability that the number of demand generated in the i-th wiring section is k-core.
[Math . C4]
(23)
$$\underset{k=0}{\overset{2F_c}{.\text{Math.}}} q_{N,k} \geq x^N \quad \text{(C4)}$$
(24) q.sub.N, k is a probability that the total of the demand number exceeding the fixed station lines in the 1st to N-th wiring sections calculated by Expression C5 is k-core.
(25)

[Math . C5]

$$(\text{Case of } N = 1)\, q_{1,k} = p_{1,(1-a)f_i + k} \quad (\text{Case of } N > 1)\, q_{N,k} = \sum_{l=0}^{k} q_{N-1,i}\, p_{N,(1-a)f_i + k - i} \qquad \text{(C5)}$$

(26) The above inventions can be combined wherever possible.

Advantageous Effects of Invention

(27) The present invention can provide an optical access network that has high reliability supporting information communication service and efficiently responds to difficult-to-predict optical fiber demand.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a diagram illustrating a configuration of an optical access network according to the present invention.

(2) FIG. **2** is a diagram for illustrating core connection at a connection point connecting an upper loop and a lower loop of the optical access network according to the present invention.

(3) FIG. **3** is a diagram illustrating an example of an area model of the optical access network according to the present invention.

(4) FIG. **4** is a diagram illustrating a loop scale dependence of an availability improvement rate calculated in the optical access network according to the present invention.

(5) FIG. **5** is a diagram illustrating the dependence of the reliability per wiring section on the number of lower loop optical fibers calculated in the optical access network according to the present invention.

(6) FIG. **6** is a diagram illustrating the flexible optical fiber ratio dependency of the number of required optical fibers of an upper loop calculated in the optical access network according to the present invention.

(7) FIG. **7** is a diagram illustrating a design method for the optical access network according to the present invention.

DESCRIPTION OF EMBODIMENTS

(8) Embodiments of the present invention will be described with reference to accompanying drawings. The embodiments described below are examples of the present invention, and the present invention is not limited to the following embodiments. In the present specification and the drawings, the components having the same reference numerals indicate the same components.

Summary of Invention

(9) FIG. **1** is a diagram illustrating a configuration of an optical access network according to the present embodiment. In addition, FIG. **2** is a diagram for illustrating optical fiber connection at a connection point connecting an upper loop and a lower loop of the optical access network according to the present embodiment. The optical access network includes: an upper loop optical fiber cable **11** which includes a plurality of fixed station lines **12** and at least one common line **13** and is laid on a loop-like path passing through a communication carrier base facility **10**, and a lower loop optical fiber cable **21** which includes at least one user line **22** and at least one redundant line **23** and is laid on a loop-like path passing through a connection point **30** with the upper loop optical fiber cable **11** and a user base facility **20**, in which a fixed connection line **32** for connecting the user lines **22** and the fixed station lines **12** with one-to-one correspondence, and an optical fiber switching function unit **31** for exclusively connecting the redundant line **23** and the common line **13** are provided at the connection point **30**, the number of common lines **13** is smaller than the total number of redundant lines **23** included in each of the plurality of lower loop optical fiber cables **21**, and the optical fiber switching function unit **31** switches connection between the redundant line **23** and the common line **13** according to demand fluctuation of the lower loop optical fiber cable **21**.

(10) A wiring section **25** is an area where the lower loop optical fiber cable **21** is laid, and a user base **20** is installed. In the present embodiment, the communication carrier base facility **10** may be described as an "exchange office **10**." The term "AVAILABILITY" is availability, and is a ratio obtained by dividing usable time (time excluding unusable time due to failure, construction, or the like) by total time.
(11) The optical access network configuration includes the optical fiber cable **11** (hereinafter, an upper loop) that is looped to connect the wiring section **25** and the exchange office **10** together, the optical fiber cable **21** (hereinafter, a lower loop) that is looped and laid in each wiring section **25**, and an optical fiber switching function unit **31** installed at the connection point **30** between the upper loop **11** and the lower loop **21**. The optical fiber switching function unit **31** is a wiring board or an optical switch which can be switched by connector connection. The wiring board or the optical switch may be remotely controllable.
(12) In this network configuration, the wiring section **25** is designed to have a size such that it touches the upper loop **11**. Then, the upper loop **11** is designed to have such a size that an availability improvement rate γ of the wiring section **25** connected to the upper loop **11** becomes a maximum. The availability improvement rate γ is obtained by the following expression, where α.sub.1 is an availability upon setting one route connecting the exchange office **10** and the wiring section **25**, and α.sub.2 is an availability upon setting two routes connecting the same.
(13) [Math. 1]

$$= \frac{_2 - _1}{_1} \quad (1)$$

(14) α.sub.1 and α.sub.2 are obtained by the following expressions.
(15) [Math. 2] [Math. 3]

$$_1 = \frac{1}{1 + -} \quad (2) \qquad _2 = \frac{1 + 2-}{1 + 2- + (-)^2} \quad (3)$$

(16) Here, λ is a failure rate indicating the number of failures per unit time, and μ is a repair rate indicating the number of repair times per unit time.
(17) Assuming that λ is proportional to the cable length and 1/μ is proportional to the number of optical fibers per cable, λ and μ are described as follows.
(18) [Math. 4] [Math. 5]

$$= C_1 r_{\text{loop}} \quad (4) \qquad \frac{1}{} = \frac{C_2 \quad (R^2 - r_{\text{loop}}^2)}{2} \quad (5)$$

(19) Here, as illustrated in FIG. **3**, a fan-shaped area having a center angle θ and a radius R with the exchange office **10** as the origin is assumed as an area shape for constructing the upper loop **11** and the wiring section **25** adjacent to the upper loop **11**. Here, r.sub.loop is a distance from the exchange office **10** to the far end of the upper loop **11**, σ is the demand density (the number of optical fibers demanded per unit area), and C.sub.1 and C.sub.2 are constants.
(20) By substituting Expressions (2) to (5) into Expression (1), the following expression is obtained as a derivation equation for the availability improvement rate.
(21) [Math. 6]

$$= \frac{C_1 C_2 \sigma\theta r_{\text{loop}} (R^2 - r_{\text{loop}}^2)}{2 + C_1 C_2 \sigma\theta r_{\text{loop}} (R^2 - r_{\text{loop}}^2)} \quad (6)$$

(22) The size of the upper loop **11** is designed based on the condition of r.sub.loop that maximizes Expression (6).
(23) The number of required optical fibers for the upper loop **11** (the total number of optical fibers included in the upper loop F.sub.total) and the number of optical fibers connected to the optical fiber switching function unit **31** (the number of common lines **13** F.sub.c) determines as follows. Considering fluctuations in the number of demand generated in the wiring section **25** (same as the above-mentioned "number of cores demanded"), the number of cores F.sub.total required to obtain a certain degree of reliability (probability of not requiring additional work) is obtained from the

conditions for minimization. In other words, since the number of optical fibers F.sub.total changes under the number of common lines Fc, the number of user lines **22** and the number of redundant lines **23** of the lower loop **21**, as shown in the FIG. **6**, the number Fc of common lines, the number of user lines **22** and the number of redundant lines **23** of the lower loop **21** are determined so that the number of cores F.sub.total is minimized.

(24) Hereinafter, a more detailed description will be given. FIG. **2** is a connection diagram of an optical fiber at a connection point between the upper loop **11** and the lower loop **21**. The optical fiber of the upper loop **11** is composed of an optical fiber (fixed station line **12**) directly connected to the lower loop **21** without being connected to the optical fiber switching function unit **31** and an optical fiber (common line **13**) connected to the lower loop **21** via the optical fiber switching function unit **31**. The common line **13** can be also connected to an optical fiber switching function unit **31** of another lower loop **21** to be shared by the plurality of lower loops **21** and to be used in any of the lower loops **21** by connection switching of the optical fiber switching function units **31**. The number of required optical fibers F.sub.total of the upper loop **11** is obtained by the following expression.

[Math. 7]

(25) 0

$$F_{\text{total}} = \sum_{i=1}^{N} (1 - a_i) f_i + F_c \quad (7)$$

(26) Here, N is the number of lower loops **21** connected to the upper loop **11**, a.sub.i is the proportion of the number of the optical fibers (redundant lines **23**) connected to the optical fiber switching function unit **31** to the required number f.sub.i of optical fibers of the i-th wiring section **25** (hereinafter, flexible optical fiber ratio), f.sub.i is the number of required optical fibers (the sum of the user line **22** and the redundant line **23**) of the lower loop **21** in the i-th wiring section **25**, and F.sub.c is the number of common lines **13**.

(27) Since the optical fiber cable laid in a loop shape can use the optical fiber from an arbitrary route in a clockwise direction and a counterclockwise direction, the demand of 2 f.sub.i cores at the maximum can be accommodated by the cable of the number of optical fibers f.sub.i in the i-th wiring section **25**. When a design value of reliability per wiring section **25** is defined as x, it is necessary to make the probability that the number of demands generated in the i-th wiring section **25** is equal to or less than 2 f.sub.i cores designed to be equal to or more than x, so that f.sub.i is designed so as to satisfy the following expression. Here, x is referred to as “reliability per wiring section.”

[Math. 8]

(28)

$$\sum_{k=0}^{2f_i} p_{i,k} \geq x \quad (8)$$

(29) Here, p.sub.i, k is the probability that the number of demand generated in the i-th wiring section **25** is k-core.

(30) On the other hand, since the F.sub.c must be designed so that the probability that the total of the number of demands exceeding the fixed station line **12** is equal to or less than 2 F.sub.c cores in the 1st to N-th wiring sections **25** is equal to or more than x.sup.N, the F.sub.c is designed so as to satisfy the following expression.

[Math. 9]

(31)

$$\sum_{k=0}^{2F_c} q_{N,k} \geq x^N \quad (9)$$

(32) Here, q.sub.N, k is the probability that the total of the demand numbers exceeding the number of fixed station lines **12** in the 1-th to N-th wiring sections **25** is k-core, and is obtained by the following expression.

[Math . 10]

(33) (Case of $N = 1$)

$$q_{1,k} = p_{1,(1-a)f_i+k}$$

(Case of $N > 1$) (10)

$$q_{N,k} = \sum_{l=0}^{k} q_{N-1,l}\, p_{N,(1-a)f_i+k-1}$$

(34) From the above, the number of required optical fibers F.sub.total of the upper loop **11** in the main network configuration is obtained so that f.sub.i (i=1 to N) obtained from Expression (8) and F.sub.c obtained from Expressions (9) and (10) are substituted into Expression (7). The number of optical fibers connected to the optical fiber switching function unit **31** of the i-th wiring section **25** is designed so that the number of optical fibers on the upper loop **11** side is 2F.sub.c and the number of optical fibers on the lower loop **21** side is a.sub.i×2f.sub.i. The flexible optical fiber ratio a.sub.i at this time is designed so as to satisfy a condition that F.sub.total becomes minimum in a.sub.i dependency of F.sub.total calculated from Expression (7).

Effects of the Invention

(35) Using a loop type wiring topology as the wiring topology, a redundant path is secured and reliability is improved, and optical fiber use efficiency is improved by sharing optical fiber resources between a clockwise route and a counterclockwise route of the loop. Further, by constituting the optical access network with a plurality of loops (an upper loop **11** and a lower loop **21**), the above-mentioned effects can be obtained at any place of the optical access network.

(36) The upper loop **11** secures a redundant configuration in a different path between the exchange office **10** and the wiring section **25**, and makes optical fiber resources shared between the wiring sections **25**. The lower loop **21** secures a redundant configuration in a different path between the connection point **30**, with the upper loop **11**, and the user base **20**, and makes optical fiber resources shared between routes in the wiring section **25**.

(37) The optical fiber switching function unit **31** has an effect of reducing required switching work operation when a demand deviation different from prediction occurs between wiring sections **25** connected to the upper loop **11**. As compared with fusion connection widely used in a conventional star-shaped network configuration, by preparing the optical fiber switching function unit **31** with a wiring board or an optical switch in advance, work operation time required for the connection switching can be reduced, and, upon allowing the wiring board or the optical switch to be remotely controlled, site work itself can be reduced.

Embodiments

(38) Embodiments of the present invention will be described in conjunction with the accompanying drawings. Here, an optical access network configuration in the area illustrated in FIG. **3** will be described as an example. It is assumed that the area shape is a fan shape having a center angle θ=π/2 radian with the exchange office **10** as an origin, and a radius R=2 km, an expected value of the number of demand generated per unit area is 100 core/km.sup.2, the number of wiring sections **25** is 4 sections, and the areas of three wiring sections **25** connected to the upper loop **11** are equal to each other. C.sub.1, C.sub.2, x are assumed as C.sub.1=10.sup.−8, C.sub.2=0.1, and x=0.9, respectively. The present invention is not limited to this, and may be used under area conditions different from those of the present embodiment.

(39) The size of the upper loop **11** is obtained from the r.sub.loop dependency of the availability improvement rate γ calculated from Expression (6). FIG. **4** shows the result of calculation of the r.sub.loop dependency of γ. As illustrated in FIG. **4**, the upper loop **11** in the present embodiment is designed to have a size such that the distance from the exchange office **10** to the far end of the upper loop **11** is 1.2 km.

(40) The number of required optical fibers f.sub.i of the lower loop **21** is designed so as to satisfy

the condition of Expression (8). In the case of r.sub.loop=1.2 km, since the area of each wiring section **25** connected to the upper loop **11** becomes 0.67 km.sup.2, the expected value of the number of demand generated in each wiring section **25** becomes 67-cores. FIG. **5** is a result of calculation of the dependence of the reliability x per wiring section **25** on the number of optical fibers of the lower loop **21** in the present embodiment. Here, a normal distribution of an expected value of 67-cores and a standard deviation of √67-cores is assumed as a probability density function of the number of demand generated per wiring section **25**. Referring to FIG. **5**, the number of required optical fibers of the lower loop **21** in the present embodiment is 56-cores.

(41) The number of required optical fibers F.sub.total and the flexible optical fiber ratio a.sub.i of the upper loop **11** are obtained from the flexible optical fiber ratio dependency of F.sub.total calculated from Expression (7). FIG. **6** shows the calculation result of the flexible optical fiber rate dependency of the number of required optical fibers of the upper loop **11** in the present embodiment. In the present embodiment, the same flexible optical fiber rate a is calculated in all the wiring sections. As illustrated in FIG. **6**, since F.sub.total converges to 156-cores of the minimum value at the flexible optical fiber ratio of equal to or more than 14%, the number of required optical fibers of the upper loop **11** in the present embodiment is 156-cores. In a case where the flexible optical fiber rate is higher than 14%, the number of required optical fibers of the upper loop **11** is not changed, and the number of connections of the optical fiber switching function unit **31** is increased. In order to avoid complication of the optical fiber switching function unit **31**, the proper value of the flexible optical fiber rate in the present embodiment is 14%.

(42) The total number of optical fibers connected to the optical fiber switching function unit **31** (the connection optical fibers **33** connected to the lower loop **11** and the clockwise and counterclockwise common lines **13** of the upper loop) is calculated as follows.

(43) The number of connected optical fibers on the upper loop **11** side (the number of clockwise and counterclockwise common lines **13** connected to the optical fiber switching function unit **31**) is calculated by following expression.

2×(number of required optical fibers of upper loop 11)−2×(1−flexible optical fiber ratio)×(number of required optical fibers of lower loop 21)×(number of wiring sections 25)=2×F.sub.total−2×(1−a.sub.i)×f.sub.i×N

(44) The number of connected optical fibers on the lower loop **21** side (the number of connected optical fibers **33** connected to the optical fiber switching function unit **31**) is calculated by following expression.

2×(flexible optical fiber ratio)×(number of required optical fibers for lower loop 21)=2×a.sub.i×f.sub.i

(45) Upon making calculation in the example described with reference to FIGS. **3** to **6**, (F.sub.total=156, a=0.14, f.sub.i=56, N=3), the number of connected optical fibers on the upper loop **11** side is 2×156−2×(1−0.14)×56×3≈24 (cores), and the number of connected optical fibers on the lower loop **21** side is 2×0.14×56≠16 (cores).

(46) In this case, the number of common lines **13** is 12 and the number of redundant lines **23** is 8.

(47) As described above, the number of connected optical fibers on the upper loop side is designed not to be equal to the total number of optical fibers connecting the plurality of lower loops and the optical fiber switching function unit, but to be smaller than the total number. For example, in the above example, since the number of connected optical fibers on the lower loop **21** side is 16-cores and the number of lower loops **21** is 3, the total number is 48-cores, but the number of connected optical fibers on the upper loop **11** side is 24-cores smaller than the total number of 48-cores. That is, the clockwise and counterclockwise 24-cores of the common line **13** of the upper loop **11** are shared by the three lower loops **21**, and each lower loop **21** can use any 16-cores of the 24-cores. This can produce a global grouping effect that the optical fiber switching function unit **31** can deal with the demand fluctuation with the number of optical fibers smaller than the total number by handling the demand fluctuation of each lower loop **21** as the demand fluctuation of the whole

three lower loops.

APPENDIX

(48) This optical access network can be designed as illustrated in FIG. **7**. (1) The optical access network includes an upper loop optical fiber cable which includes a plurality of fixed station lines and at least one common line and is laid on a loop-like path passing through a communication carrier base facility, and a lower loop optical fiber cable which includes at least one user line and at least one redundant line and is laid on a loop-like path passing through a connection point with the upper loop optical fiber cable and a user base facility, in which a fixed connection line for connecting the user line and the fixed station lines in a 1:1 ratio, and an optical fiber switching function unit for exclusively connecting the redundant line and the common line are provided at the connection point, the number of the common lines is smaller than the total number of the redundant lines included in each of the plurality of lower loop optical fiber cables, and the optical fiber switching function unit switches connection between the redundant line and the common line according to demand fluctuation of the lower loop optical fiber cable.

(49) The optical access network is designed by performing steps S01 and S02. (2) In step S01, the loop-like path of the upper loop optical fiber cable is a path in which an improvement rate γ of availability in a case of the upper loop optical fiber cable is maximized with respect to availability when there is one path from the communication carrier base facility to the connection point in a wiring section where the lower loop optical fiber cable is laid. (3) Specifically, the improvement rate γ expressed by Expression C1, and the loop-like path of the upper loop optical fiber cable is r.sub.loop that maximizes the improvement rate γ.

[Math. C1]

(50)
$$= \frac{C_1 C_2 \sigma\theta r_{\text{loop}}(R^2 - r_{\text{loop}}^2)}{2 + C_1 C_2 \sigma\theta r_{\text{loop}}(R^2 - r_{\text{loop}}^2)} \quad \text{(C1)}$$

(51) Where an area shape constructed by the upper loop optical fiber cable is a fan shape having a center angle θ and a radius R with the communication carrier base facility as an origin, the r.sub.loop is a distance from the communication carrier base facility to the connection point, σ is a demand density, and C1 and C2 are constants. (4) In step S02, the number of optical fibers connecting the redundant line and the optical fiber switching function unit is a number that converges the number of optical fibers F.sub.total included in the upper loop optical fiber cable to a minimum. (5) Specifically, the optical fiber number F.sub.total is calculated by Expression C2:

[Math. C2]

(52)
$$F_{\text{total}} = \sum_{i=1}^{N} (1 - a_i) f_i + F_c \quad \text{(C2)}$$

where N is the number of the lower loop optical fiber cables connected to the upper loop optical fiber cable, a.sub.i is the proportion of the number of the redundant lines among the optical fibers included in the i-th lower loop optical fiber cable (hereinafter, flexible optical fiber ratio), and f.sub.i is an optical fiber included in the i-th lower loop optical fiber cable, and Fc is the number of the common lines.

(53) Here, f.sub.i satisfies Expression C3, and Fc satisfies Expression C4.

[Math. C3]

(54)
$$\sum_{k=0}^{2f_i} p_{i,k} \geq x \quad \text{(C3)}$$

(55) x is a design value of reliability per wiring section where the lower loop optical fiber cable is laid, and p.sub.i, k is a probability that the number of demand generated in the i-th wiring section is k-core.

[Math. C4]

(56)
$$\sum_{k=0}^{2F_c} q_{N,k} \geq x^N \quad \text{(C4)}$$

(57) q.sub.N, k is a probability that the total of the demand number exceeding the fixed station lines in the 1st to N-th wiring sections calculated by Expression C5 is k-core.

[Math . C5]

(58)

$$\text{(Case of } N = 1) \quad q_{1,k} = p_{1,(1-a)f_i + k} \quad \text{(Case of } N > 1) \quad q_{N,k} = \sum_{l=0}^{k} q_{N-1,l}\, p_{N,(1-a)f_i + k - 1} \tag{C5}$$

REFERENCE SIGNS LIST

(59) **10**: Communication carrier base facility (exchange office) **11**: Upper loop optic fiber cable **12**: Fixed station line **13**: Common line **20**: User base facility **21**: Lower loop optical fiber cable **22**: User line **23**: Redundant line **25**: Wiring section **30**: Connection point **31**: Optical fiber switching function unit **32**: Fixed connection line **33**: Connection optical fiber

Claims

1. An optical access network comprising: an upper loop optical fiber cable which includes a plurality of fixed station lines and at least one common line and is laid on a loop-like path passing through a communication carrier base facility; and a lower loop optical fiber cable of a plurality of lower loop optical fiber cables which includes at least one user line and at least one redundant line and is laid on a loop-like path passing through a connection point with the upper loop optical fiber cable and a user base facility, wherein a fixed connection line for connecting the user line and the fixed station lines in a 1:1 ratio, and an optical fiber switching function unit for exclusively connecting the redundant line and the common line are provided at the connection point, the number of the common lines is smaller than the total number of the redundant lines included in each of the plurality of lower loop optical fiber cables, and the optical fiber switching function unit switches connection between the redundant line and the common line according to demand fluctuation of the lower loop optical fiber cable.
2. The optical access network according to claim 1, wherein the loop-like path of the upper loop optical fiber cable is a path in which an improvement rate γ of availability in a case of the upper loop optical fiber cable is maximized with respect to availability when there is one path from the communication carrier base facility to the connection point in a wiring section where the lower loop optical fiber cable is laid.
3. The optical access network according to claim 2, wherein the improvement rate γ expressed by Expression C1, and the loop-like path of the upper loop optical fiber cable is r.sub.loop that maximizes the improvement rate γ:

[Math . C1]

$$= \frac{C_1 C_2 \sigma\theta r_{loop}(R^2 - r_{loop}^2)}{2 + C_1 C_2 \sigma\theta r_{loop}(R^2 - r_{loop}^2)} \tag{C1}$$

where an area shape constructed by the upper loop optical fiber cable is a fan shape having a center angle θ and a radius R with the communication carrier base facility as an origin, the r.sub.loop is a distance from the communication carrier base facility to the connection point, σ is a demand density, and C1 and C2 are constants.
4. The optical access network according to claim 1, wherein the number of optical fibers connecting the redundant line and the optical fiber switching function unit is a number that converges the number of optical fibers F.sub.total included in the upper loop optical fiber cable to a minimum.
5. The optical access network according to claim 4, wherein the optical fiber number F.sub.total is

calculated by Expression C2: [Math. C2]

$$F_{\text{total}} = \sum_{i=1}^{N} (1 - a_i) f_i + F_c \quad \text{(C2)}$$

where N is the number of the lower loop optical fiber cables connected to the upper loop optical fiber cable, a.sub.i is the proportion of the number of the redundant lines among the optical fibers included in the i-th lower loop optical fiber cable (hereinafter, flexible optical fiber ratio), and f.sub.i is an optical fiber included in the i-th lower loop optical fiber cable, and Fc is the number of the common lines, here, f.sub.i satisfies Expression C3, and Fc satisfies Expression C4, [Math. C3]

$$\sum_{k=0}^{2f_i} p_{i,k} \geq x \quad \text{(C3)}$$

x is a design value of reliability per wiring section where the lower loop optical fiber cable is laid, and p.sub.i, k is a probability that the number of demand generated in the i-th wiring section is k-core, [Math. C4]

$$\sum_{k=0}^{2F_c} q_{N,k} \geq x^N \quad \text{(C4)}$$

q.sub.N, k is a probability that the total of the demand number exceeding the fixed station lines in the 1st to N-th wiring sections calculated by Expression C5 is k-core, [Math. C5]

$$\begin{aligned} &(\text{Case of } N = 1) \\ &q_{1,k} = p_{1,(1-a)f_i + k} \\ &(\text{Case of } N > 1) \\ &q_{N,k} = \sum_{l=0}^{k} q_{N-1,l}\, p_{N,(1-a)f_i + k - 1} \end{aligned} \quad \text{(C5)}$$

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